Do Not Let English Judge the World: Protocol Sensitivity in Multilingual LLM-as-a-Judge Evaluation

Anonymous authors
Paper under double-blind review

Abstract

LLM-as-a-judge evaluation is increasingly used to score open-ended model outputs, but multilingual benchmarks often hide an implementation choice that can change conclusions: which language the judge sees, which language the rubric uses, and whether non-English examples are translated into English before judging. We study this protocol sensitivity on multilingual summarization evaluation using SEAHORSE human labels and source-grounded XLSum examples, with a small WMT MQM translation-quality contrast. Across 400 candidate-quality examples in English, Spanish, Turkish, and Vietnamese, the same `gpt-4o-mini` judge assigns materially different scores under direct English rubrics, target-language rubrics, explicit English-pivot judging, and bilingual rubrics. The largest shifts appear in target-language form dimensions: for Vietnamese grammar, target-language rubric and English-pivot scores differ by 1.20 points on a 1–5 scale on average. English pivot is not a safe default: on Turkish comprehensibility it significantly underperforms direct judging, and a stronger `gpt-4.1-mini` audit reproduces the same failure mode. In source-grounded XLSum main-ideas evaluation, protocol shifts are smaller but remain language-dependent. In WMT MQM translation quality, reference-based direct-vs-pivot deltas are smaller, while reference-free judging exposes larger `gpt-4o-mini` protocol effects and a targeted stronger-judge audit shows which effects are directional versus model-dependent. Across 17 non-audit task/language cells, the best protocol is not direct in 11 cells and explicit pivot is worst in 10. We argue that multilingual benchmarks should report protocol sensitivity, language gaps, calibration behavior, parse rates, and cost instead of a single aggregate judge score.

1 Introduction

LLM judges are attractive because they make open-ended evaluation cheap, fast, and easy to scale. Prior work has shown that LLM judges can correlate with human preferences and task-specific human ratings in settings such as chatbot evaluation, summarization, and machine translation (Zheng et al., 2023; Liu et al., 2023; Kocmi & Federmann, 2023). This has made LLM-as-a-judge a common evaluation component in research and product pipelines.

Multilingual evaluation adds a protocol choice that is often treated as incidental. A benchmark builder can ask an English prompt to judge a non-English answer directly, translate the answer into English first, translate the rubric into the target language, or provide bilingual instructions. These choices are not equivalent. English-pivot judging may make semantic content easier for an English-dominant model to process, but it can erase target-language form evidence such as grammar, fluency, register, morphology, and repetition.

This paper asks a practical question: when a multilingual benchmark reports an LLM judge score, how sensitive is that score to the judge protocol? We study this question using SEAHORSE multilingual human ratings (Clark et al., 2023), source-grounded XLSum examples (Hasan et al., 2021), and a reference-based WMT MQM translation-quality contrast. Our contribution is a small but reusable protocol comparison package:

- A benchmark scaffold covering 580 main base examples across candidate-quality, source-grounded semantic, and reference-based translation-quality settings, plus a 200-item stronger-judge audit.
- A direct comparison of four multilingual judge protocols for summarization, plus reference-based and reference-free WMT MQM translation-quality extensions.
- Protocol-sensitivity metrics: by-language human alignment, paired score shifts, language gaps, calibration behavior, parse rate, and cost.
- A Global Judge Reporting Card for future multilingual LLM-as-a-judge papers.

Our main finding is not that one protocol is universally best. The key result is that protocol choice can change both scores and human alignment, and the direction depends on language and quality dimension.

2 Related Work and Positioning

LLM judges are now widely studied as scalable evaluators for open-ended generation, including chatbot response quality, summarization, and machine translation (Zheng et al., 2023; Liu et al., 2023; Kocmi & Federmann, 2023). This line of work establishes that prompted LLM evaluators can align with human judgments in some settings, but it also raises concerns about prompt sensitivity and evaluator bias. Our focus is narrower: we treat the judge protocol itself as an experimental factor in multilingual evaluation.

Recent multilingual judge work finds that LLM judges can be inconsistent across languages and that disagreement can be exploited or corrected by more elaborate multilingual judging procedures (Fu & Liu, 2025; Fu & Lu, 2026). Our contribution is complementary. Rather than proposing a new judge model or adjudication algorithm, we evaluate low-cost reporting choices available to benchmark builders: judging original text, translating into English, translating the rubric, using bilingual instructions, and calibrating score thresholds against small human-labeled splits.

3 Experimental Setup

Candidate-quality setting. We sample from the SEAHORSE test split. The main run uses four languages: English (`en-US`), Spanish (`es-ES`), Turkish (`tr`), and Vietnamese (`vi`). We evaluate two candidate-visible dimensions: comprehensibility and grammar. For each language and dimension, we sample 50 balanced examples, with 25 positive and 25 negative human labels. This gives 400 base examples and 1,600 judge responses across four protocols.

Source-grounded semantic setting. SEAHORSE raw TSV files do not include source documents. To recover source-grounded examples, we match SEAHORSE reference summaries to raw XLSum test summaries. Matching recovers 151/151 Spanish, 243/244 Turkish, and 252/252 Vietnamese reference rows. We then sample 30 balanced main-ideas examples per language, giving 90 base examples and 360 judge responses.

WMT MQM translation-quality contrast. To test whether protocol sensitivity is equally severe for machine translation evaluation, we sample 30 balanced examples per language pair from WMT MQM human-evaluation rows: `en-de`, `en-ru`, and `zh-en`. We use year-2022 rows and derive high/low labels from within-pair MQM-score quantiles. This gives 90 base examples. We run both reference-based judging (source, candidate, reference) and reference-free judging (source, candidate only). Each condition uses direct judging and explicit English-pivot judging for all pairs, plus target-language and bilingual rubrics for the two non-English target pairs, giving 600 WMT judge responses total.

Stronger-judge audit. To check whether the observed protocol sensitivity is only a cheap-judge artifact, we run `gpt-4.1-mini` on a stratified candidate-quality subset: 25 examples per language/dimension, 200 base examples, and 800 judge responses.

Table 1: Dataset sampling audit for paper-facing item sets.

Run	Items	Langs	Dims	Pos	Neg	Source
candidate n50	400	4	2	200	200	0%
candidate audit n25	200	4	2	96	104	0%
semantic n30	90	3	1	45	45	100%
wmt n30 shared items	90	3	1	45	45	100%
wmt ref-free audit zh-en	30	1	1	15	15	100%
wmt ref-free audit en-de	30	1	1	15	15	100%

Table 2: Prompt inventory for paper-facing runs.

Run	Prompts	Protocols	Langs	Dims	Judge
candidate n50	1600	4	4	2	gpt-4o-mini
candidate audit n25	800	4	4	2	gpt-4.1-mini
semantic n30	360	4	3	1	gpt-4o-mini
wmt reference n30	300	4	3	1	gpt-4o-mini
wmt ref-free n30	300	4	3	1	gpt-4o-mini
wmt ref-free audit	120	3	2	1	gpt-4.1-mini

Sampling audit. Table 1 summarizes the processed item sets used by the paper-facing runs. The main candidate-quality, semantic, and WMT item pools are balanced by construction, while the candidate audit intentionally has 12 positive and 13 negative examples per language/dimension cell.

Protocols. We compare four prompt surfaces:

- P0: original candidate summary with English instructions.
- P1: original candidate summary with a target-language rubric.
- P2: English translations of source text or reference text, when present, and candidate summary.
- P3: bilingual English and target-language rubric wording.

The summarization settings use all four protocols. The WMT MQM contrast uses P0 and P2 for all pairs, and P1/P3 for non-English target pairs where target-language rubric wording is meaningful. All calls request strict JSON with a 1–5 score, coarse label, confidence, and short rationale.

Prompt auditability. Table 2 summarizes the prompt files used by each paper-facing run. The release package also includes a redacted prompt inventory that preserves the system messages, rubric wording, protocol instructions, and JSON contracts while replacing item-specific source, reference, and candidate text with redaction markers.

Metrics. We report Spearman correlation, AUROC, mean absolute paired score shift, language gap, post-hoc threshold calibration, parse rate, and approximate API cost from returned usage metadata. Paired protocol comparisons use bootstrap intervals for item-level AUROC deltas.

Table 3: Candidate-quality protocol sensitivity. Protocol columns report Spearman / AUROC.

Cell	Direct	Target	Pivot	Bilingual	Comparison	ΔAUROC	95% CI	Shift
tr / comprehensibility	0.603 / 0.836	0.556 / 0.810	0.331 / 0.686	0.575 / 0.822	Pivot - direct	-0.150	[-0.257, -0.055]	0.620
vi / comprehensibility	0.349 / 0.690	0.164 / 0.590	0.086 / 0.548	0.399 / 0.714	Bilingual - pivot	0.166	[0.009, 0.317]	0.960
vi / grammar	0.231 / 0.622	0.254 / 0.628	0.016 / 0.509	0.098 / 0.550	Pivot - target	-0.119	[-0.268, 0.025]	1.200

Table 4: Stronger-judge audit on a stratified n=25-per-cell subset.

Cell	Direct	Pivot	Bilingual	Pivot-Direct ΔAUROC	Bilingual-Pivot ΔAUROC
tr / comprehensibility	0.729 / 0.910	0.168 / 0.593	0.796 / 0.946	-0.317	0.353
vi / grammar	0.564 / 0.814	-0.052 / 0.471	0.491 / 0.769	-0.343	0.298

4 Results

4.1 Protocol Choice Changes Candidate-Quality Scores

Table 3 summarizes the strongest candidate-quality findings. Protocol shifts are large on the same item, especially for target-language form dimensions. For Vietnamese grammar, target-language rubric and English-pivot scores differ by 1.20 points on a 1–5 scale on average. For Vietnamese comprehensibility, bilingual and English-pivot scores differ by 0.96 points on average.

These shifts are not merely score-scale differences. In Turkish comprehensibility, direct judging reaches Spearman 0.603 and AUROC 0.836, while explicit English pivot drops to Spearman 0.331 and AUROC 0.686. The paired AUROC delta for pivot minus direct is -0.150, with 95% bootstrap CI [-0.257, -0.055]. Thus a common workaround, translating into English before judging, can significantly reduce alignment with human labels.

Qualitative examples suggest two concrete mechanisms. In one Vietnamese human-negative comprehensibility item, target-language judging scores the original summary 2, while the English pivot "Prepare materials." is scored 5 because the translation is clear but has removed the target-language error signal. Conversely, in a Turkish human-positive comprehensibility item, direct, target-language, and bilingual protocols all score 5, but the pivot translation repeats the sentence and the pivot judge assigns 1. These cases illustrate why translation can both clean away errors and introduce new artifacts.

4.2 A Stronger Judge Does Not Remove the Failure Mode

The `gpt-4.1-mini` audit is still bounded, so we do not treat its exact correlations as final. It is useful as a stronger-model failure-mode check. On Turkish comprehensibility, the stronger judge gives direct judging Spearman 0.729 / AUROC 0.910 and bilingual judging 0.796 / 0.946, but explicit pivot falls to 0.168 / 0.593. The paired AUROC delta for pivot minus direct is -0.317, and bilingual minus pivot is +0.353. The same pattern appears for Vietnamese grammar: direct judging reaches 0.564 / 0.814, explicit pivot drops to -0.052 / 0.471, and bilingual judging recovers to 0.491 / 0.769.

4.3 Source-Grounded Semantics Are Also Protocol-Sensitive

The source-grounded main-ideas setting gives a different but complementary pattern. Protocol shifts are smaller than for grammar/form dimensions, which is expected: semantic content can often survive translation better than grammar or fluency. Still, explicit pivot is not uniformly best.

For Spanish main ideas, explicit pivot performs best. For Turkish, bilingual judging is best. For Vietnamese, all protocols are weak, but explicit pivot is chance-level. Paired comparisons reinforce this language dependence: for Turkish, bilingual minus pivot AUROC delta is +0.124 with 95% CI [0.000, 0.275]; for Vietnamese, bilingual minus pivot AUROC delta is +0.107 with 95% CI [0.013, 0.220].

Table 5: Source-grounded XLSum main-ideas alignment by protocol. Values are Spearman / AUROC.

Language	Best	Best Sp/AUROC	Direct	Pivot	Bilingual
es-ES	P2_explicit_pivot	0.514 / 0.771	0.488 / 0.762	0.514 / 0.771	0.432 / 0.724
tr	P3_bilingual	0.588 / 0.793	0.463 / 0.738	0.339 / 0.669	0.588 / 0.793
vi	P3_bilingual	0.195 / 0.607	0.143 / 0.576	0.000 / 0.500	0.195 / 0.607

Table 6: WMT MQM translation-quality protocol extension. Values are Spearman / AUROC.

Setting	Pair	Best	Best Sp/AUROC	Direct	Target	Pivot	Bilingual	Pivot-Direct ΔAUROC
Reference	en-de	P3_bilingual	0.471 / 0.689	0.388 / 0.673	0.318 / 0.596	0.398 / 0.678	0.471 / 0.689	0.004
Reference	en-ru	P1_target_rubric	0.320 / 0.649	0.165 / 0.573	0.320 / 0.649	0.135 / 0.564	0.159 / 0.576	-0.009
Reference	zh-en	P0_direct_english	0.354 / 0.693	0.354 / 0.693	n/a	0.304 / 0.669	n/a	-0.024
Ref-free	en-de	P3_bilingual	0.677 / 0.838	0.511 / 0.736	0.479 / 0.740	0.437 / 0.709	0.677 / 0.838	-0.027
Ref-free	en-ru	P1_target_rubric	0.199 / 0.600	0.023 / 0.511	0.199 / 0.600	0.035 / 0.518	0.178 / 0.589	0.007
Ref-free	zh-en	P0_direct_english	0.595 / 0.820	0.595 / 0.820	n/a	0.187 / 0.600	n/a	-0.220

4.4 Reference-Based MT Is a Useful Contrast

The WMT MQM contrast is intentionally smaller and narrower than the summarization runs, but it is important for claim boundaries. Reference-based translation-quality judging shows smaller direct-vs-pivot shifts and no significant paired AUROC deltas. For `en-de`, direct and pivot judging are nearly identical: direct Spearman/AUROC is 0.388 / 0.673 and pivot is 0.398 / 0.678. For `en-ru`, both direct and pivot are weak. For `zh-en`, pivot shifts more scores than the other pairs, with mean absolute shift 0.53, but the paired AUROC delta remains small at -0.024 with 95% CI [-0.227, 0.177].

Reference-free MT judging surfaces stronger `gpt-4o-mini` protocol effects. For `zh-en`, direct judging reaches 0.595 / 0.820, while explicit pivot drops to 0.187 / 0.600; the paired pivot-minus-direct AUROC delta is -0.220 with 95% CI [-0.440, -0.007]. For `en-de`, bilingual reference-free judging is best at 0.677 / 0.838 and improves over pivot by +0.129 AUROC with CI [0.025, 0.258]. For `en-ru`, the target-language rubric is best by AUROC, though all correlations remain low. However, a targeted `gpt-4.1-mini` WMT audit separates directional robustness from model dependence: it preserves the `zh-en` direct-over-pivot direction (0.503 / 0.773 vs. 0.254 / 0.636; delta -0.138, CI [-0.346, 0.067]) but does not reproduce the `en-de` bilingual-over-pivot gain (pivot 0.563 / 0.784 vs. bilingual 0.503 / 0.760). Thus English pivoting is risky for target-language form and some source-grounded semantic settings, but the WMT evidence is best treated as protocol/model-sensitivity evidence rather than a stable global protocol ranking.

4.5 No Single Protocol Dominates

Table 8 aggregates the per-cell instability rather than treating any one protocol as a winner. Across 17 non-audit task/language cells, the best AUROC protocol is not direct in 11 cells, explicit pivot is the worst protocol in 10 cells, and 6 cells have at least one paired AUROC delta with a bootstrap CI excluding zero. The stronger audits show the same qualitative point while also tightening the WMT claim boundary: protocol choice is an experimental factor to report, not a nuisance prompt detail to hide.

4.6 Repeatability Control

The large protocol shifts are not explained by ordinary run-to-run variation in exact repeated prompts. The historical n=20 pilot and the n=50 main run overlap on 42 original-text prompt calls with identical prompt text. Across these repeated calls, exact score agreement is 92.9%, mean absolute score delta is 0.071, and the maximum score delta is 1. This is much smaller than the main protocol shifts, such as 0.96 for Vietnamese comprehensibility and 1.20 for Vietnamese grammar. By contrast, 14 repeated explicit-pivot prompt IDs have different prompt text because the English-pivot pipeline was regenerated; these have lower exact agreement and mean absolute delta 0.643. We therefore treat the clean exact-prompt result as

Table 7: Targeted WMT reference-free stronger-judge audit with `gpt-4.1-mini`. Values are Spearman / AUROC.

Cell	Comparison	Protocol A	A Sp/AUROC	Protocol B	B Sp/AUROC	ΔAUROC	95% CI	Shift
zh-en ref-free	Pivot - direct	Direct	0.503 / 0.773	Pivot	0.254 / 0.636	-0.138	[-0.346, 0.067]	0.667
en-de ref-free	Bilingual - pivot	Pivot	0.563 / 0.784	Bilingual	0.503 / 0.760	-0.024	[-0.086, 0.016]	0.200

Table 8: Protocol-instability summary by run. AUROC range is best minus worst protocol within a language/dimension cell.

Run	Cells	Median range	Max range	Range $\geq$ .10	Best $\neq$ direct	Pivot worst	Sig. cells	Max shift
candidate n50	8	0.074	0.166	3/8	4/8	4/8	2/8	1.200
semantic n30	3	0.107	0.124	2/3	3/3	2/3	1/3	0.700
wmt reference n30	3	0.084	0.093	0/3	2/3	2/3	0/3	0.533
wmt ref-free n30	3	0.129	0.220	2/3	2/3	2/3	3/3	0.700
candidate audit n25	8	0.162	0.353	7/8	5/8	4/8	4/8	1.160
wmt ref-free audit	2	0.081	0.138	1/2	1/2	1/2	0/2	0.667

a judge repeatability control and the explicit-pivot result as evidence that translation-pivot workflows add another source of pipeline volatility.

4.7 Aggregate Scores Hide Language Gaps

Protocol choice changes language disparity. In candidate-quality comprehensibility, explicit pivot has a Spearman language gap of 0.625, from Vietnamese 0.086 to English 0.711. Bilingual judging reduces the same gap to 0.302. In source-grounded main ideas, explicit pivot again has the largest language gap, 0.514, from Vietnamese 0.000 to Spanish 0.514. In WMT, the reference-free direct protocol has a much larger gap than the reference-based direct protocol, 0.572 versus 0.223.

4.8 Calibration Is Useful but Not Universal

Post-hoc threshold calibration behaves differently across settings. In candidate-quality n=50, mean held-out balanced-accuracy delta is -0.017, with median 0.000. In the stronger audit, it remains slightly negative: mean -0.029, median -0.003. In source-grounded main ideas, calibration is positive: mean +0.136, median +0.136. In the WMT reference-based extension it is again negative, with mean -0.036 and median 0.000. A repeated learning-curve analysis gives the same boundary: with 50 stratified resamples, source-grounded semantics improves by about +0.12 to +0.14 balanced accuracy even with small calibration sets, while candidate-quality calibration remains slightly negative on average and reference-free WMT is mixed.

Table 13 explains the calibration mechanism. The source-grounded semantic judge scores are compressed low: with balanced human labels, the fixed score ≥ 4 threshold marks only 8.3% of examples as good, and the modal best in-group threshold is 2. WMT reference-free scores are compressed in the opposite direction: score ≥ 4 marks 90.0% of examples as good, and the modal best threshold is 5. Calibration is therefore useful as a score-threshold diagnostic, not a universal repair for protocol sensitivity.

5 Global Judge Reporting Card

We recommend that multilingual LLM-as-a-judge papers report: judge model and version; prompt/rubric language; whether source, candidate, or reference texts were translated; whether target-language form dimensions were judged in the original language; per-language human alignment; protocol-shift metrics on the same items; language gaps for each protocol; calibration split size and held-out effect; parse failure rate; returned token usage, and approximate cost per 1,000 judgments. Token usage is the stable accounting quantity; dollar values in our tables are returned usage estimates computed with the run-time local price map and should be refreshed against current provider pricing before

Table 9: Repeatability control from overlapping pilot and main-run caches.

Control	Pairs	Identical prompts	Exact agree	Mean $\|\Delta\|$	Max $\|\Delta\|$
exact original-text prompt repeat	42	100.0%	92.9%	0.071	1
explicit-pivot pipeline repeat	14	0.0%	57.1%	0.643	4

Table 10: Largest cross-language Spearman gaps.

Run	Dimension	Protocol	Gap	Min	Max
candidate n50	comprehensibility	P2_explicit_pivot	0.625	vi (0.086)	en-US (0.711)
candidate n50	comprehensibility	P1_target_rubric	0.535	vi (0.164)	en-US (0.699)
candidate n50	grammar	P3_bilingual	0.456	tr (-0.072)	en-US (0.385)
semantic n30	main_ideas	P2_explicit_pivot	0.514	vi (0.000)	es-ES (0.514)
semantic n30	main_ideas	P3_bilingual	0.393	vi (0.195)	tr (0.588)
semantic n30	main_ideas	P1_target_rubric	0.357	vi (0.138)	tr (0.495)
wmt n30	translation_quality	P3_bilingual	0.312	en-ru (0.159)	en-de (0.471)
wmt n30	translation_quality	P2_explicit_pivot	0.264	en-ru (0.135)	en-de (0.398)
wmt n30	translation_quality	P0_direct_english	0.223	en-ru (0.165)	en-de (0.388)
wmt n30	translation_quality	P1_target_rubric	0.002	en-de (0.318)	en-ru (0.320)
wmt ref-free n30	translation_quality_ref_free	P0_direct_english	0.572	en-ru (0.023)	zh-en (0.595)
wmt ref-free n30	translation_quality_ref_free	P3_bilingual	0.499	en-ru (0.178)	en-de (0.677)
wmt ref-free n30	translation_quality_ref_free	P2_explicit_pivot	0.402	en-ru (0.035)	en-de (0.437)
wmt ref-free n30	translation_quality_ref_free	P1_target_rubric	0.280	en-ru (0.199)	en-de (0.479)

camera-ready release. For this study, the release package includes a reproducibility manifest with file hashes, record counts, and validation commands for the processed items, response caches, analysis tables, figures, and manuscript artifacts, plus claim-evidence and RQ-coverage matrices mapping each headline claim and planned contribution to concrete artifacts and validator checks.

6 Limitations

The current evidence is deliberately bounded. All main judge calls use OpenAI models, and the stronger audits use another OpenAI model rather than an independent model family. Target-language rubric translations are pragmatic pilot translations, not native-speaker validated. The source-grounded setting covers one semantic dimension and three non-English languages. The WMT MQM contrast is small, covers one year and three language pairs, adds target/bilingual rubrics only for the two non-English target pairs, and uses quantile-derived binary labels from continuous MQM scores. The targeted WMT stronger-judge audit covers only two reference-free cells at n=30, so it is a robustness check rather than a definitive model-family comparison. The reference-free WMT condition reuses the same MQM-derived labels, so it is a protocol stress test rather than a new human annotation target. SEAHORSE labels are binary yes/no labels for each dimension, while LLM judges produce 1–5 scores; thresholding and calibration are therefore part of the evaluation design.

7 Conclusion

Multilingual LLM-as-a-judge evaluation is under-specified without the judge protocol. The same item can receive different scores depending on whether the judge sees original text, translated text, target-language instructions, or bilingual instructions. English-pivot judging can help in some semantic and reference-based settings, but it can also erase target-language evidence and increase language gaps. We recommend that global benchmark builders report protocol sensitivity as a first-class evaluation result.

Table 11: Held-out threshold calibration effects.

Run	Groups	Mean ΔBalAcc	Median ΔBalAcc
candidate n50	32	-0.017	0.000
audit n25	32	-0.029	-0.003
semantic n30	12	0.136	0.136
wmt n30	10	-0.036	0.000
wmt ref-free n30	10	0.059	0.000
wmt audit zh-en	2	-0.045	-0.045
wmt audit en-de	2	0.000	0.000

Table 12: Repeated threshold-calibration learning curve. Entries are mean held-out balanced-accuracy deltas over repeated stratified calibration samples.

Run	$n_{cal} = 2$	$n_{cal} = 8$	Largest n_{cal}	Largest Δ	P(improve)	P(degrade)
candidate n50	-0.054	-0.022	24	-0.009	0.198	0.331
semantic n30	0.127	0.120	24	0.136	0.612	0.112
wmt reference n30	-0.067	-0.047	24	-0.041	0.000	0.180
wmt ref-free n30	0.013	0.055	24	0.076	0.404	0.136
candidate audit n25	-0.066	-0.032	16	-0.024	0.232	0.351

Table 13: Score-threshold diagnostic. Rates use the fixed score ≥ 4 baseline threshold.

Run	Groups	Pos score	Neg score	Pred good	Pos good	Neg good	Mode best t
candidate n50	32	3.55	2.78	48.3%	62.6%	33.9%	4
semantic n30	12	2.23	1.76	8.3%	10.0%	6.7%	2
wmt reference n30	10	3.95	3.55	73.3%	84.0%	62.7%	4
wmt ref-free n30	10	4.70	4.19	90.0%	96.0%	84.0%	5
candidate audit n25	32	4.15	3.03	58.0%	76.8%	40.6%	4

Table 14: Run inventory, parse rate, and observed incremental API cost.

Run	Base	Calls	Parse	Cost	Cost/1k
candidate n50	400	1600	100.0%	0.1114	0.070
candidate audit n25	200	800	100.0%	0.1431	0.179
semantic n30	90	360	100.0%	0.0659	0.183
wmt reference n30	90	300	100.0%	0.0295	0.098
wmt ref-free n30	90	300	100.0%	0.0206	0.069
wmt ref-free audit	60	120	100.0%	0.0221	0.184

Table 15: Price-independent API usage inventory from returned token metadata. API calls include translation calls and retries recorded in the run cost paths.

Run	API calls	Input tok.	Output tok.	Total tok.	Tok./call
candidate n50	2000	382769	89897	472666	236.3
candidate audit n25	1005	198162	48377	246539	245.3
semantic n30	450	252403	46780	299183	664.9
wmt reference n30	390	108238	22157	130395	334.3
wmt ref-free n30	300	84237	13339	97576	325.3
wmt ref-free audit	120	32659	5649	38308	319.2

Table 16: Observed API cost estimates from returned usage metadata.

Run	Approx. cost
candidate n50	0.1114
audit n25	0.1431
semantic n30	0.0659
wmt n30	0.0295
wmt ref-free n30	0.0206
wmt ref-free audit	0.0221

A Figures

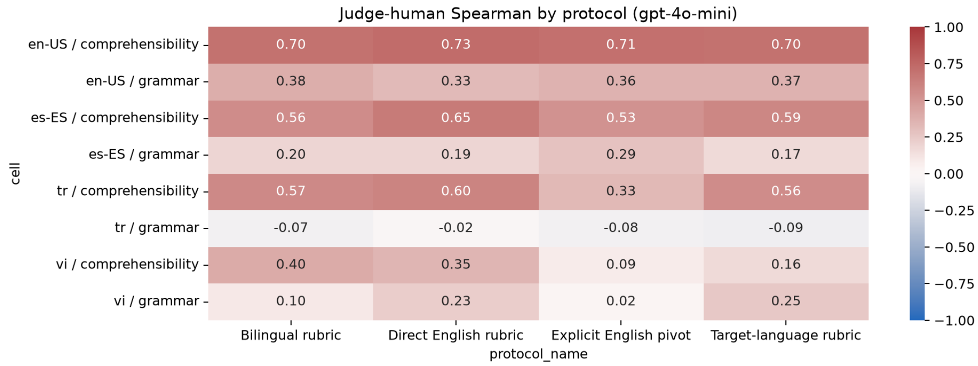

Figure 1: Candidate-quality n=50 judge-human Spearman by language, dimension, and protocol.

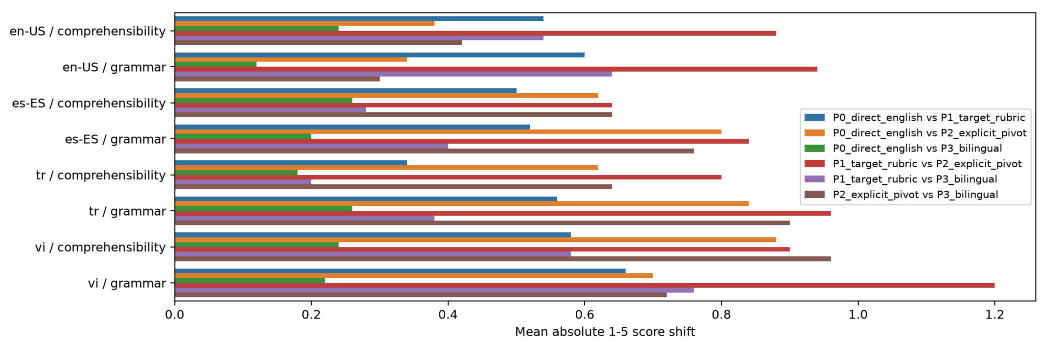

Figure 2: Candidate-quality n=50 same-item protocol shifts.

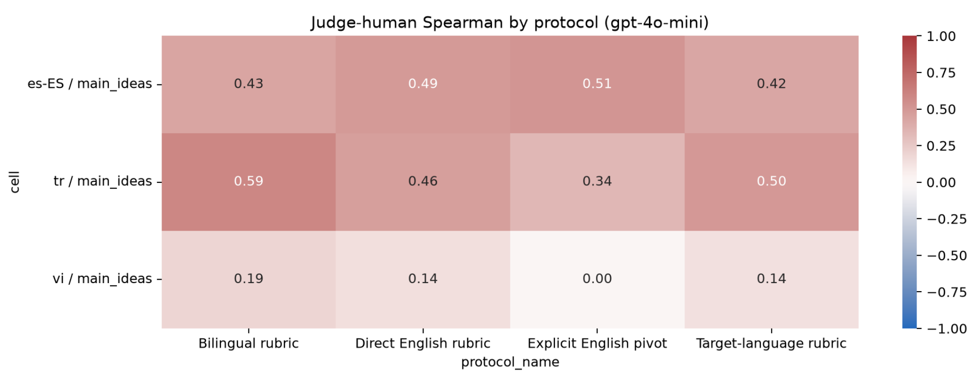

Figure 3: Source-grounded semantic n=30 judge-human Spearman by language and protocol.

References

Elizabeth Clark, Shruti Rijhwani, Sebastian Gehrmann, Joshua Maynez, Roee Aharoni, Vitaly Nikolaev, Thibault Sellam, Aditya Siddhant, Dipanjan Das, and Ankur P. Parikh.

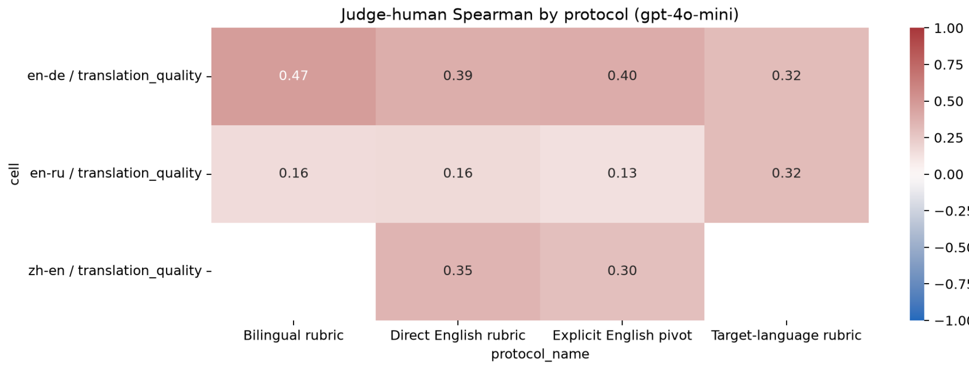

Figure 4: WMT MQM n=30 translation-quality judge-human Spearman by language pair and protocol.

Seahorse: A multilingual, multifaceted dataset for summarization evaluation. *arXiv preprint arXiv:2305.13194*, 2023.

Xiyan Fu and Wei Liu. How reliable is multilingual LLM-as-a-judge? *arXiv preprint arXiv:2505.12201*, 2025.

Xiyan Fu and Wei Lu. When languages disagree: Self-evolving multilingual LLM judges. *arXiv preprint arXiv:2606.08092*, 2026.

Tahmid Hasan, Abhik Bhattacharjee, Md. Saiful Islam, Kazi Mubasshir, Yuan-Fang Li, Yong-Bin Kang, M. Sohel Rahman, and Rifat Shahriyar. XL-Sum: Large-scale multilingual abstractive summarization for 44 languages. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, 2021.

Tom Kocmi and Christian Federmann. Large language models are state-of-the-art evaluators of translation quality. In *Proceedings of the Eighth Conference on Machine Translation*, 2023.

Yang Liu, Dan Iter, Yichong Xu, Shuohang Wang, Ruochen Xu, and Chenguang Zhu. G-Eval: NLG evaluation using GPT-4 with better human alignment. *arXiv preprint arXiv:2303.16634*, 2023.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-judge with MT-bench and chatbot arena. *arXiv preprint arXiv:2306.05685*, 2023.